FINE-GRAINED GARMENT PARSING: A BODY GENERATION APPROACH

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ABSTRACT

Current human parsing methods segment an image into different semantic parts including background, body parts and garments. A major limitation of today's human parsing methodologies is that they are not able to provide fine-grained garment segmentation (e.g., left and right sleeves), and it is mainly due to the lack of a dataset with such fine-grained semantic garment part labels. To tackle this, we propose a body generation approach. In particular, we first use a body generation module based on image inpainting, to locate the fine-grained garment parts corresponding to where the generated body parts are, e.g., the left sleeve is assumed to be associated with the left arm; we then extract the garment parts from the original *whole* garment based on the positions above. In our experiments based on a public dataset focusing on top clothing images, our solution can effectively separate a top garment into a left sleeve, a right sleeve and front, as compared to state-of-the-art solutions that parse it as a whole.

Index Terms— garment parsing, fine-grained, body generation

1. INTRODUCTION

Recent years have witnessed increasing adoption of deep learning in fashion tasks, including fashion trends prediction [1], outfit or garment recommendation [2], as well as computer-aided clothes design [3] and virtual try-on systems [4, 5]. In these tasks, understanding the semantic information of fashion images plays a non-trivial role, and fashion image parsing represents one of the most important semantic information extraction techniques [5]. However, existing human parsing methods segment the garments in a fashion image as a whole, and cannot parse fine-grained garment parts. As illustrated in Fig. 1, the existing method parses an upper clothing as a whole while our method parses it into a left sleeve, a right sleeve and front.

To predict fine-grained fashion image parsing results, an intuitive idea is to improve the existing coarse-grained human parsing methods [6], e.g., by training their models on a fine-grained garment segmentation dataset. However, to the best of our knowledge, there does not exist such dataset with fine-grained garment segmentation labels. As creating such a dataset is time-consuming in practice given the massive combinations of fashion items, in this paper, we propose to predict fine-grained garment parsing with only the original fashion images.

Fig. 1. **Current human parsing and our fine-grained garment parsing.** The last two columns show the results generated by existing parsing method and our method. Our model can predict the left and right sleeve segmentation from the upper clothing.

Our idea is to first use a *body generation* method that "removes" the garment from the body, to locate the positions of the body parts (e.g., left arm) as an assistance to infer the location of the garment parts (e.g., left sleeve), and then accomplish the fine-grained garment parsing. Though the idea seems straightforward, we are still facing great challenges: how to generate realistic body images, especially when there

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are not sufficient paired training samples for naked-body images and wrapped-body images in practice, and how to leverage the generated body images to produce the garment segmentation?

Our answers to these questions are a design of a weakly-supervised garment parsing method, which involves two stages. The first body generation stage generates the body parts (usually wrapped by the garments in the original image); and the second fine parsing stage gives the final garment segmentation based on the corresponding garment part locations inferred from the previous body part locations. The whole solution and our contributions are summarized as follows.

First, we propose a weakly-supervised method based on image *inpainting* [7] to generate naked human body parts. To tackle the challenges of unpaired naked and wrapped images, 1) we propose to use only the images with naked body parts (e.g., sleeveless images for arms) in the inpainting model training process; 2) we further mask the selected images with the coarse-grained garments segmentation that can be predicted by existing methods. Using this weakly supervision, we tend to enable the model to generate the naked body parts, which are called *generated-body image* in our design, under the input of both naked-body and wrapped-body samples.

Second, we propose to infer the location of a garment part using the location of the corresponding body part generated above, and perform the parsing. In particular, we use the pre-trained human parsing model Graphonomy [6] on both the original images and the generated-body images; then we extract the fine-grained garment segmentation by computing difference-sets between the human maps of the generated-body image and the human maps of the original image. Experimental results on a public top clothing dataset demonstrate the effectiveness of our two-stage model.

2. RELATED WORK

Human Parsing. Human parsing is to decompose a human image into semantic fashion/body regions. Some works solve the human parsing task as a general semantic segmentation task. For example, Chen et al. [8] used dilated convolution to increase the receptive field effectively, and Liang et al. [9] used the LSTM network to inherently combine local and global spatial dependencies. A more effective and more targeted approach was to use human pose estimation to provide high-level semantic information for human parsing in [10, 11]. In [6], Gong et al. investigated universal human parsing via graph transfer learning and introduced a new universal human parsing agent which was named Graphonomy. Some well-labeled datasets, such as PASCAL-Person-Part [12] , ATR [13] , CIHP [14] , LIP [11] , also provide strong support for these work. However, in all previous human parsing tasks, a garment is considered as a whole and there is no dataset annotated with fine-grained garment parts. In this paper, we propose a human body generation method to help predict a finer segmentation of the garment without the need for corresponding ground truth labels.

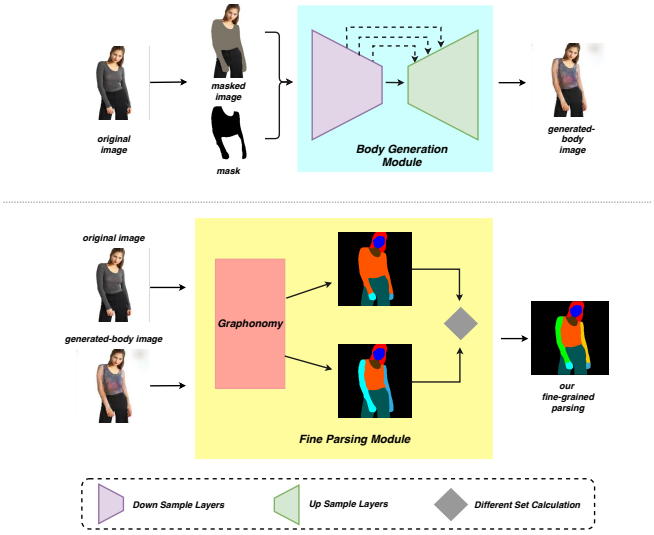


Fig. 2. An overview of our fine-grained garment parsing method. The structure consists of two main modules: (a) Body Generation Module: to generate naked human arms from the masked input. (b) Fine Parsing Module: to predict the sleeve segmentation via the intermediate body generation.

Image Inpainting. Image inpainting is a work that fills in the holes in an image, and has a lot of applications. For example, it can be used in image editing to generate a plausible image region when moving unwanted image content. Some non-learning methods of image inpainting recover the target region from neighboring pixels using some mechanisms such as distance field [15, 16] . Patch-based methods ([17, 18] recover the image by searching for relevant patches from the non-hole regions or other source images. PatchMatch [17] speeds it up by proposing a faster patch searching algorithm.

With the development of deep learning, other methods [19, 20, 21] use convolution networks to recover the image. However, these approaches focus on rectangular-shaped holes inpainting, but fail to handle irregular masks. Liu et al. [7] introduced a large irregular mask dataset and proposed the use of partial convolutions for image inpainting on irregularly shaped holes. In our work, we use the image inpainting method based on partial convolution to generate the skin texture in the original wrapped body region, which assists in predicting the garment part segmentation corresponding to this body region.

3. METHODOLOGY

3.1. Overview

We design a two-stage structure via body generation to parse the garments into fine-grained semantic parts. The structure includes two modules, one is a Body Generation Module (BGM), and the other is a Fine Parsing Module (FPM). Firstly, in BGM, we use an image inpainting network to generate the human body part from an image with the mask of corresponding garment part. Then in FPM, we get the parsing results of original image (wrapped body part by the corresponding garment parts) and generated-body image (naked body part by "removing the garment parts") via a pre-trained universal parsing model. Finally, we predict the garment part segmentation by computing the difference sets between these two parsing results in FPM. The overall pipeline is illustrated in Fig. 2.

3.2. Body Generation Module

Our base model is an image inpainting model [7], which can fill in the holes in the images. Our target is to "restore" naked body part by masking the corresponding garment (which wraps partial region of the targeted generated body part) and body region (which is the unwrapped region of the targeted generated body part) in the human image.

Mask. It is easy to restore the naked body part which is not wrapped by garments in a human image, while it is difficult to restore it from an image with the body part occluded by the garment in virtue of lacking its naked counterpart. So in order to make the model treat images of different wrapped-body levels equally as naked-body images, we mask both the garments which bring occlusion and the unwrapped body part from the human segmentation map. The mask method is to set the pixels of corresponding garment and body part as 0, and reserve the other original pixels, as the column 'input' in Fig. 3 shown.

Architecture. The backbone of our network is PConvNet [7] . We adopt U-Net [22] structure for the encoder-decoder body generator design, which uses skip connection to share information between layers. We replace the convolution manipulation of our network by partial convolution like PConvNet.

The operation of the ordinary convolution layer is:

$$F^{out} = W * F^{in} + b \tag{1}$$

where $*$ is the convolution operation, W is the parameter of convolution kernel and b is the bias. F^{in} denotes the feature map of input and F^{out} denotes the feature map of output.

Compared to the ordinary convolution, partial convolution only extracts the valid pixels in the feature map by using a mask with the same size as the feature map. This mask is binary and 1 means that the corresponding pixel is valid and 0 means the contrary. At the beginning of a partial convolution operation, we use an all-1 convolution kernel W_1 with the same size as the ordinary convolution kernel W to do convolution operation with the current mask M^{old}, and get M^{sum}:

$$M^{sum} = W_1 * M^{old} \tag{2}$$

Then we use M^{sum} to update the mask:

$$M^{new}_{ijk} = \begin{cases} 1, & if \quad M^{sum}_{ijk} > 0 \\ 0, & otherwise \end{cases} \tag{3}$$

where (i,j) is the position of the pixel and k is the index of the clannel.

Then, to ensure the extraction of valid pixels, we use the mask M^{old} to weigh the feature map F^{in} firstly and thus manipulate the convolution operation:

$$C = W * (F^{in} \circ M^{old}) \tag{4}$$

where $\circ$ is Hadamard product and C denotes the feature map of output under the above weighted feature map of input.

Finally, to keep consistency with the mask and adaptively adjust the value of the feature map according to the number of valid pixels, partial convolution adopts a new feature map update operation. If the value in M^{new} is 0, we set the value of the same position in the feature map as zero, and for the others, we use M^{sum} to scale the feature map:

$$F^{new}_{ijk} = \begin{cases} C_{ijk} \Big/ M^{sum}_{ijk} + b, & if \quad M^{new}_{ijk} > 0 \\ 0, & otherwise \end{cases} \tag{5}$$

where F^{new} is the updated feature map.

3.3. Fine Parsing Module

In fact, the output of the body generation module is not the real human body image. It is to generate the similar texture as human skin at the location of the wrapped body part. So for the wrapped-body cases, the body generation output is "naked human body part with the contour of original garments there ". The generated body region can be predicted as the human body by the current parsing methods through our experiments. That is to say, for the wrapped-body images, human parsing methods can label the body region generated by BGM as body parts.

Based on this insight, we introduce our design to achieve garment part segmentation from the whole garment. We leverage the relationship between the garment part and its wrapped body part, e.g., the regions of the generated arms are inherently corresponding to the regions of the original sleeves. The general method is to compute the difference-sets

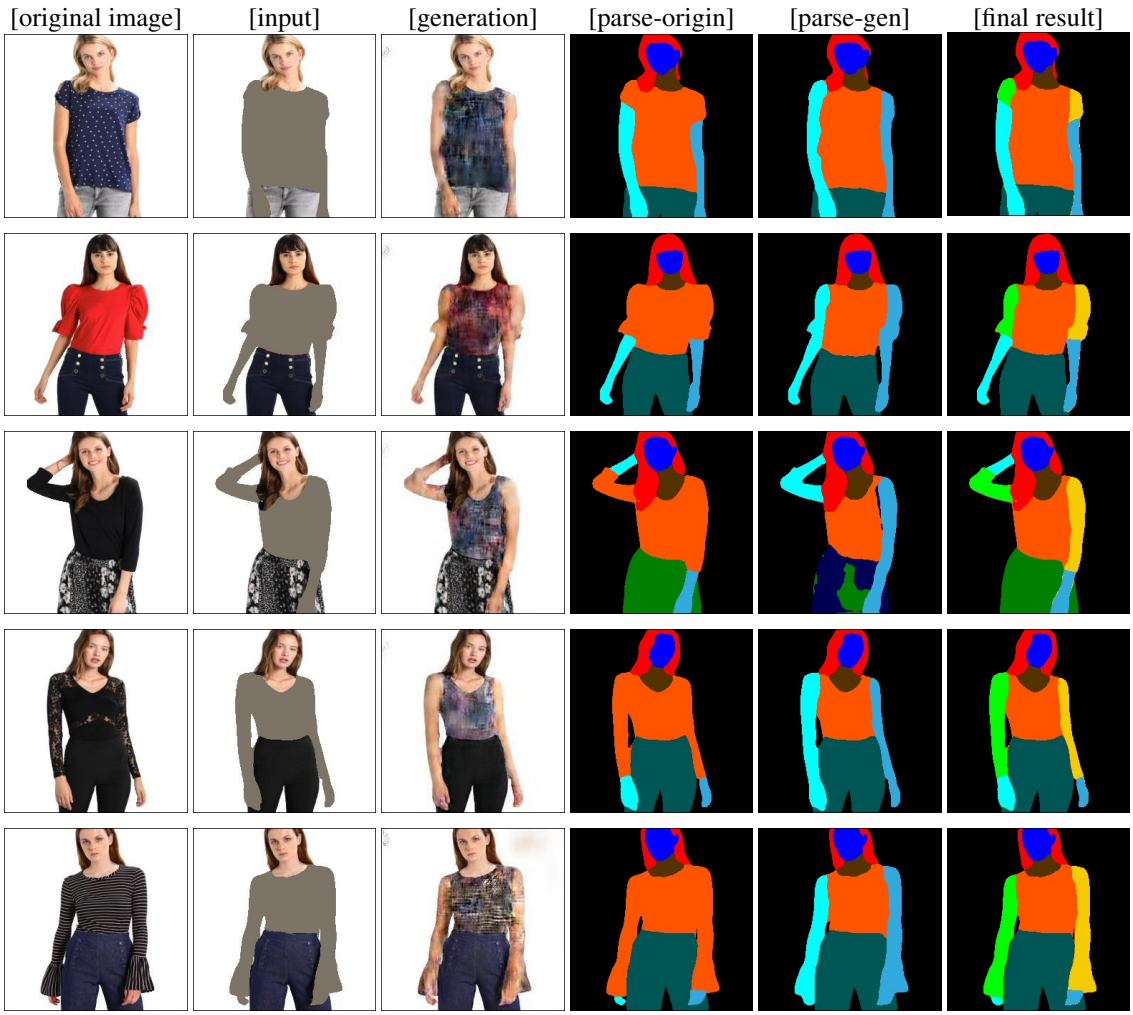


Fig. 3. **Visualised output of different steps in our method.** 'generation' shows the intermediate output of the body generation module. 'parse-origin' and 'parse-gen' are the parsing results of the original image and intermediate generation output via Graphonomy.

between the human maps of generated-body image and original human image. Firstly, we choose Graphonomy [6] pre-trained on Pascal-Person-Part, ATR and CIHP as our parsing model. Then we predict the segmentation of the original image and generated image via Graphonomy, and regard these two human maps as two *sets*, denoted as S_{origin} and S_{BGM}. At last, we subtract S_{origin} from S_{BGM} to segment the corresponding garment part S_{g-part} in the original image:

$$S_{g-part} = S_{BGM} - S_{origin} \tag{6}$$

4. EXPERIMENTS

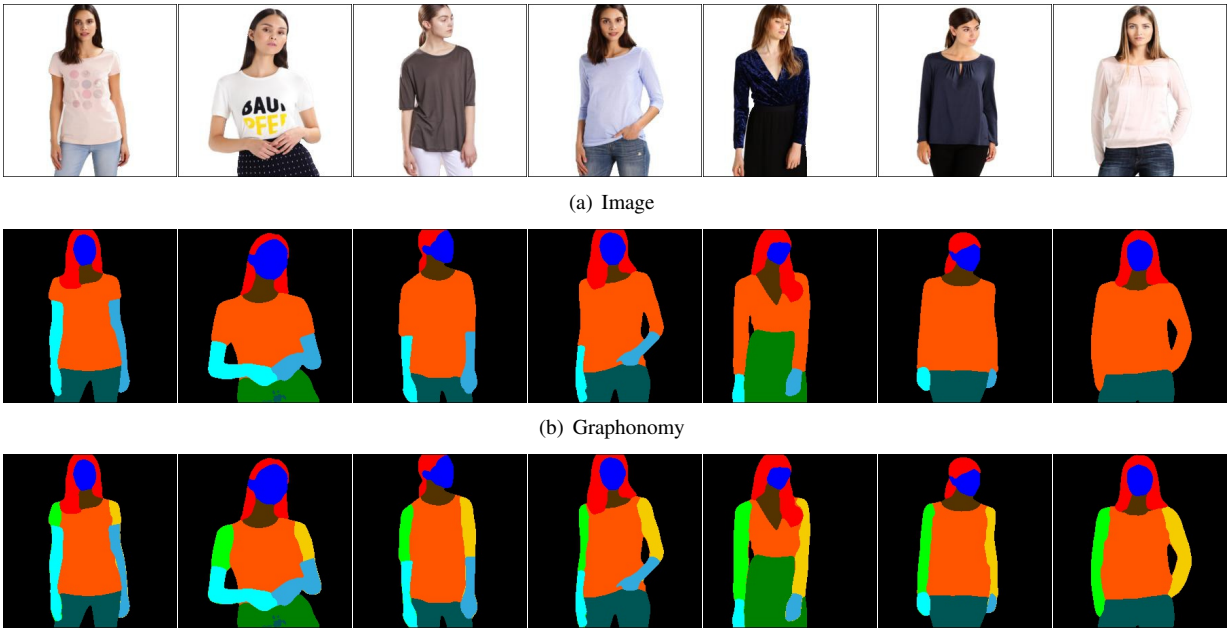
In this work, we focus on the fine-grained parsing of top garments. Our goal is to predict the segmentations of sleeves and front from a whole top garment.

4.1. Dataset Processing

We use the data from a virtual try-on dataset in [5], which focuses on the top clothing worn by models. We use a sleeve-length classifier pre-trained on FashionAI [23] to pick out the sleeveless top clothes images in the original train data as our train dataset. The other data is used as test dataset, because we want to predict the sleeve segmentation at various levels of sleeve-length. The size of the train dataset is 2298 and the size of the test dataset is 13955.

4.2. Implementation Details

Training Setup. The loss we use for body generation module is the same as in [7], which is a combination of l_1 loss L_{valid}



(a) Image

(b) Graphonomy

(c) Ours

Fig. 4. **Performance comparison with state-of-the-art method.**

and L_{hole} for valid pixels and masked pixels, perceptual loss $L_{perceptual}$, style loss L_{style} and total variation loss L_{tv}. We use Adam for optimization and a fixed learning rate of 0.0002. We train body generation module on a single GTX 1080-Ti GPU (11GB) with a batch size of 16 for $500K$ iterations. All input images are resized to 256×256 and the output images have the same resolution.

Body Generation Module. We use a 14-layer U-Net with seven 2-strided down-sampling convolutional layers and seven up-sampling layers. Like regular U-Net design [5] , we use the combination of nearest-neighbor interpolation layers and 1-strided convolutional layers to replace 2-strided deconvolution layers which are usually used for up-sampling. Besides, we use partial convolution operation to take the place of all ordinary convolution operation. The numbers of filters for down-sampling convolutional layers are 64, 128, 256, 512, 512, 512, 512. The numbers of filters for up-sampling convolutional layers are 512, 512, 512, 256, 128, 64, 3. Except for the first down-sampling and last up-sampling convolutional layers, each convolutional layer is followed by the Batch-Normalization layer. The nonlinearity operation for down-sampling convolutional layers is ReLU. And the nonlinearity operation for up-sampling convolutional layers (except the last up-sampling convolutional layer, which uses the nonlinearity of ReLU) is LeakyReLU with the scope set as 0.2.

Fine Parsing Module. In the parsing inference stage, we do the same operation as [6]. In order to stabilize the parse prediction, we average results of left-right flipped images and multi-scale inputs from 0.5 to 1.75 with an increment of 0.25.

4.3. Analysis and Comparison

Analysis of our method. Fig. 3 presents visual outputs of different steps using our two-stage method. It can be seen that our body generation module performs effective in assisting in parsing the region of the sleeves. The body generation module virtually generates the skin texture in the region of the sleeves (including the original arms region for they are all masked as the input of this module). The generation outputs of the third and fourth row show that human arms can be generated well by our body generation module. Even though the worse or terrible intermediate body generation happens, it will not have a big effect on our parsing result. The body generation of the first and second row generates the skin texture in the region covering the original big sleeve region, which looks like artifical human arms. But through our fine parsing module, the fine-level parsing result transfer the incompatible generated arms into the realistic sleeve segmentation. Similarly, in the last row, the intermediate body generation wrongly preserves the original sleeves and makes the body generation weird. The following fine parsing module eliminates this problem by extracting the different part sets between the original parsing output and body generation parsing output. Therefore, our two-stage model is effective in parsing realistic and reasonable sleeves from the top clothes.

Comparison with state-of-the-art. Since our proposed fine-grained segmentation is a new task, we choose one of the current state-of-the-art methods as comparative method to show the effectiveness of our method. We test different sleeve-

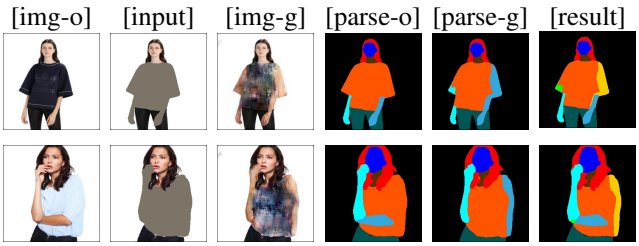


Fig. 5. **Failure cases of our method.** 'img-o' represents the original images, 'img-g' represents the generated images by the body generation module, and 'parse-o' and 'parse-g' are the parsing outputs of original images and generated images.

length fashion images on our method and Graphonomy [6]. The comparative results are shown in Fig. 4. It is obvious that current parsing methods can just parse the top garments as a whole while our method can segment the left and right sleeves from the top clothing. More importantly, our method is suitable for all fashion images with a variety of sleeve lengths and diverse poses.

4.4. Failure Cases

There are still some failure cases in our work. Sample results are illustrated in Fig. 5. One of the failure reasons is that body generation module cannot work correctly when the inner border of the top clothing after masked is hard to detect (as the first case shown). The other is that when the pose of human is rare and exists occlusion, it is hard to locate the arms location in the 'input' for body generation module (as the second case shown).

5. CONCLUSION

To tackle the problem of fine-grained garment parsing for fashion images, we introduce a two-stage body generation approach. The body generation module is designed to locate the garment positions for the corresponding body parts, and the parsing module predicts the garment part segmentation by computing the difference-sets between human maps of the original image and generated-body image. Experimental results verify the effectiveness of our study.

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